



Nonstoichiometric tungsten oxide: structure, synthesis, and applications

Lu Zhang¹ · Hao Wang¹ · Jingbing Liu¹ · Qianqian Zhang¹ · Hui Yan¹

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Abstract

Stoichiometric tungsten oxide (WO_3) and tungsten bronzes have received much attention due to the energy-related functional performances. In addition, as a transition metal oxide, substoichiometric tungsten oxides have manifested their unique properties and exceptional potential to lots of applications; however, no other studies have concentrated on the oxygen-deficient tungsten oxides till now. Herein, this article presents the origin of structure and synthesis of $\text{W}_{18}\text{O}_{49}$ nanomaterials chiefly, and then the general introduction of different applications, including electrochemistry, photocatalysis, near-infrared shielding and phototherapy and sensing.

1 Introduction

Tungsten oxides (WO_x , $x \leq 3$), or $\text{WO}_{3-\delta}$ ($\delta \geq 0$) is a transition metal oxide with wide-ranging applications, early the interest in WO_x dated back to the tungsten bronze type of Cs_xWO_3 . Tungsten bronzes, a group of nonstoichiometric compounds with a general formula of M_xWO_3 (or $\text{M}_x^+\text{W}_{1-x}^{6+}\text{W}_x^{5+}\text{O}_3$) $\text{M} = \text{Na}^+$, K^+ , Rb^+ , Cs^+ , NH_4^+ ($0 < x < 1$), have been identified as functional materials. More recently, a substantial number of researches about WO_3 were stimulated by the discovery of electrochromic. WO_3 is constituted by perovskite structure and is one of the promising candidates for photocatalysis, but also electrochemistry, thus it can be regarded as a better electrode for electrochromic and supercapacitors [1–4].

Several substoichiometric WO_x have been experimentally demonstrated. They are known as Magnéli phases, which are characterized by formation of WO_7 pentagonal bipyramids surrounded by five edge-sharing W–O octahedral. Extensive experimental evidence for the effect of the oxygen vacancy to electrochromic properties is investigated. It is observed that WO_x films exhibit diverse properties [5, 6]: $x < 2.5$ films are metallic and conductive, $x = 2.7$ – 2.5 films are blue

and conductive, and as for $x > 2.7$ films are transparent and resistive.

In the case of WO_x , a variety of oxygen-deficient tungsten oxides can be obtained, e.g., $\text{WO}_{2.72}$ ($\text{W}_{18}\text{O}_{49}$), $\text{WO}_{2.8}$ (W_5O_{14}), $\text{WO}_{2.83}$ ($\text{W}_{24}\text{O}_{68}$) [7, 8], $\text{WO}_{2.9}$ ($\text{W}_{20}\text{O}_{58}$) [9], $\text{WO}_{2.94}$ [10], $\text{W}_{19}\text{O}_{55}$ [11]. Additionally, monoclinic $\text{W}_{18}\text{O}_{49}$ has been reported that can be isolated in a pure form [12]. Apart from the strong absorption in the UV region [13], $\text{W}_{18}\text{O}_{49}$ nanocrystals show a broad absorption in both visible and near-infrared (NIR) region, which leads to the possible development of the applications such as light harvesting in photovoltaics, cancer photothermal therapy, bioimaging, and biosensing.

In the past years, many significant achievements based on stoichiometric WO_3 and tungsten-based compounds including tungsten bronze oxide have been obtained. Herein, in this article, we describe the fundamental crystal structures of nanostructured WO_x , and focus on the nonstoichiometric WO_x , particularly on $\text{W}_{18}\text{O}_{49}$. Meanwhile, we summarize the different reported synthesis methods of $\text{W}_{18}\text{O}_{49}$ nanostructures, finally review their interesting applications including photocatalysis, electrochromic, gas sensors, and cancer therapy that exploit this function-oriented material $\text{W}_{18}\text{O}_{49}$.

2 Structure

In this section, we discuss the structure of nonstoichiometric WO_x —tungsten oxides (WO_x) are a series of oxides where the lattice can withstand substantial oxygen deficiency, or

✉ Jingbing Liu
liujingbing@bjut.edu.cn

¹ The College of Materials Science and Engineering,
Beijing University of Technology, Beijing 100124,
People's Republic of China

called at mixed valence state tungsten-based compounds (W^{3+} , W^{4+} , W^{5+} , W^{6+}). The crystal structure of nonstoichiometric WO_x is described in detail, particularly the better known nonstoichiometric compositions $W_{18}O_{49}$.

At an earlier time, it was found increasingly that Wadsley has termed “grotesque stoichiometry” in place of nonstoichiometry [14], attributed to the advent of electron microscopy particularly, and therefore, researchers paid more attention on “extended defects” and aimed to construct a framework employed in the area relating structure and stoichiometry. What is called crystallographic shear (CS) has drawn a great deal of attention to nonstoichiometric oxides as the principal structural motif [15, 16]. Researchers revealed a great variation in structural types among the CS phases already known [17]. Magnéli phase is of a homologous series of phases, each with extremely small deviations from its stoichiometric composition, a nuanced distinction of oxygen element concentration. It is found that lots of structures can be built up by combining metal–oxygen octahedra (MX_6 octahedra) in a similar way with regard to some tungsten oxides [18], such as W_nO_{3n-2} [19] and W_nO_{3n-1} [20].

In substoichiometric oxides, the structure element may contain two kinds of polyhedra, viz., MX_6 octahedra, and MX_7 pentagonal bipyramids that share its equatorial edges with five MX_6 to form units of the kind (Fig. 1a), conveniently called pentagonal columns (PCs) [16]. In reduced tungsten trioxide, another structural defect involving PCs has been observed, hexagonal tunnels (HT). As is shown in Fig. 1b, doublets of edge-linked PCs are joined by diamond links to form infinite slabs, which are mutually connected via corner-sharing WO_6 octahedra. In complicated structure of $W_{18}O_{49}$, it could be considered as alternatively buildup of stacked PC-HT-PC units. This should be obvious from Fig. 1c. The existence of these structures based upon the

ReO_3 structure established resulted in the nature of defects, centered largely upon the slightly reduced transition metal oxides, notably $WO_{3-\delta}$, $TiO_{2-\delta}$, and $MoO_{3-\delta}$ [18].

In a word, the crystallography of substoichiometric oxides could be depicted as construction of metal–oxygen octahedra joined by corners, which extend infinitely in two dimensions and have a finite in a third direction, finally the formation of ReO_3 -type blocks. The blocks are mutually joined along some given planes by octahedra having edges in common.

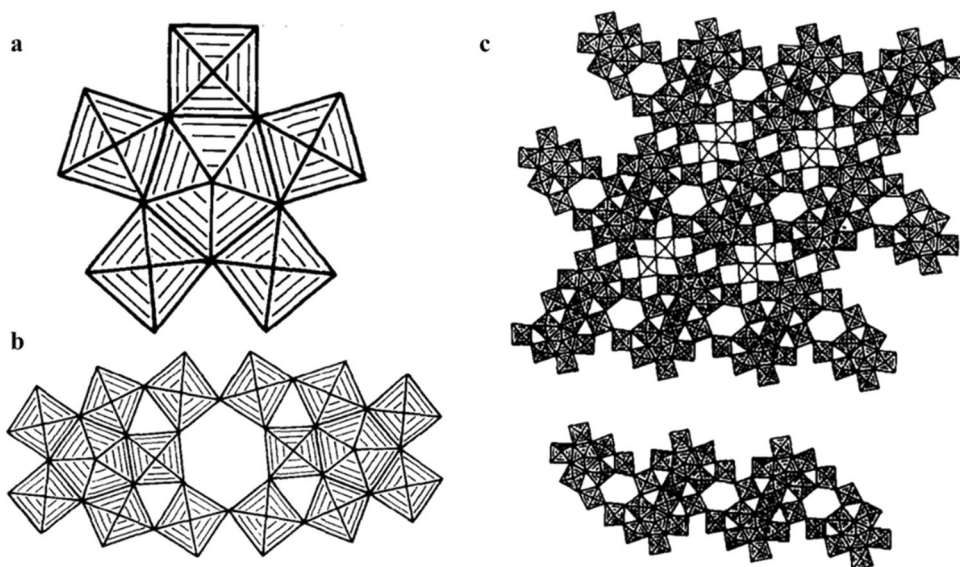
3 Synthesis

In this section, we discuss different approaches for the syntheses of nanostructured WO_x , which implemented classified roughly vapor- and liquid-phase methods. Solution-based synthesis, or called liquid-phase synthesis methods, such as templating, sol–gels, hydrothermal or solvothermal, electrodeposition, is facile and common for the preparation of WO_x with various nanomorphologies, e.g., nanowires, nanorods [21, 22], nanobundles, hollow spheres, nanodots [23].

Tungsten source materials like tungsten hexacarbonyl ($W(CO)_6$), tungsten hexachloride (WCl_6), usually dissolve in organic solvents, and then, the mixture is heated at high temperature for some time, as the organic solvents partially reduce W^{6+} to W^{5+} , the obtained product is separated by centrifugation and washed from the organic reactants.

Park et al. [24] reported a simple preparation of tungsten oxide nanorods by solution-based colloidal approach. A mixture of $W(CO)_6$, trimethylamine *N*-oxide (Me_3NO), oleylamine was stirred in three-neck flask with Schlenk line technique. A rodlike morphology (aspect ratio ≈ 20) with average diameter (4 ± 1 nm) and length (75 ± 20 nm) is observed. Similarly Grauer [25] and co-workers have

Fig. 1 **a** Pentagonal column (PC). **b** PC-HT-PC group. **c** Structures of and $W_{18}O_{49}$. Reproduce with permission [16]. Copyright 1982 Published by Elsevier Inc

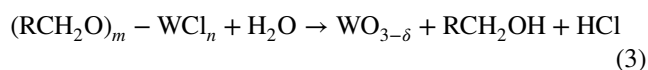
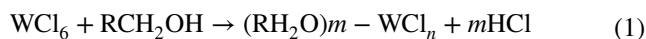


synthesized nanorods with $W(CO)_6$ and mixture of trimethylamine *N*-oxide and various amine, e.g., octadecylamine, hexadecylamine, tetradecylamine, dodecylamine in 1-octadecene (ODE) by heating in an oil bath. Moshofsky [26] and co-workers demonstrated a new method for WO_x nanowires (NWs) with independently control of the length and width. The ultrathin nanowire is 1.2 ± 0.2 nm in diameter and hundreds of nanometers in length. By varying the reaction conditions it can be given different lengths (10–700 nm) and diameters (1–4 nm).

WCl_6 is another raw material to synthesis the nanostructures of $W_{18}O_{49}$. Colloidal WO_x nanowires are synthesized by high temperature non-hydrolytic (solvothermal) reaction [27]. The diameter of each individual nanowire is approximately 4 nm on average, while about 250 nm in average length. Guilian Li and co-workers have synthesized $W_{18}O_{49}$ nanostructures (rod, bundle, and fibers) via a high-temperature but short-time process [28]. The bundle-like nanostructures were obtained when halve the concentration of WCl_6 in the starting solution, and end up with nanorods for double the WCl_6 . And the $W_{18}O_{49}$ NWs were synthesized by using ammonium metatungstate hydrate (AMT, $(NH_4)_6H_2W_{12}O_{40} \cdot xH_2O$) and oleylamine [29, 30]. The wire-like $W_{18}O_{49}$ was characterized by transmission electron microscopy (TEM), wherein nanowires ~ 30 nm in length and ~ 2.5 nm in diameter were found. $WO_{3-\delta}$ nanorods were prepared by thermal decomposition process with slight modification reference to the above mentioned. Typically, AMT, oleylamine, and nonanoic acid were added into the reaction mixture [31]. WCl_4 also can be the raw material for synthesis of $W_{18}O_{49}$ [32]. $W_{18}O_{49}$ nanorod (NR) was synthesized by thermally induced crystal growth process. The TEM image indicates that the as-synthesized $W_{18}O_{49}$ is a monodispersed NR with ~ 4 nm width and ~ 50 nm length [33].

3.1 Solvothermal

Solvothermal is simple and scalable for fabrication of nanoarchitectures, such as nanowires [34, 35], nanorods [36, 37], and hollow sphere. In a typical procedure, certain amount of $W(CO)_6$, or WCl_6 , as raw materials, was dissolved in absolute alcohols solution, and then, the mixture was sealed in a Teflon-lined autoclave and heated at some temperature for a period of time. A precipitate was collected by centrifugation and repeatedly washed with organic solvents (e.g., ethanol or others) and distilled water, etc., followed by vacuum drying. The nanostructures generally form by the following reactions:



Guo, C [38] prepared the blue powder of $W_{18}O_{49}$ sample consists of homogenous nanospheres with diameter 300–400 nm; however, the surfaces of these spheres are covered with many nanorods, while on the whole it exhibits an urchin-like morphology. Similarly, Jung [39] synthesized $W_{18}O_{49}$ consists of entangled nanowires with a length of up to 300 nm, which form a 3-D porous structure. The high-resolution transmission electron microscopy (HR-TEM) image reveals that the diameter of each nanowire is about 7 nm. And scanning electron microscopy (SEM) and TEM images show the as-synthesized $W_{18}O_{49}$ by alcoholthermal method has urchin-like structure assembled between nanowires with diameter less than 20 nm and length (500–600 nm) [40].

In situ growth of $W_{18}O_{49}$ nanocrystal has become an efficient routine to the film preparation incorporated with the solvothermal method. In the previous literature, Liu et al. [41] have prepared ultrathin $W_{18}O_{49}$ nanowires using a modified solvothermal approach. A fluorine-doped tin oxide (FTO) glass was placed into an autoclave loaded with the solution, while the other condition still unchanged. The as-synthesized sample was comprised of ultrathin $W_{18}O_{49}$ nanowires with the diameters of ~ 18 nm. Besides that, in another paper, the as-grown $W_{18}O_{49}$ was composed of entangled nanowires with several micrometers in length and width of 10–20 nm [42]. Also, other substrate could be placed into the reactor during the process of solvothermal. A supercapacitor electrode based on nanowire $WO_{2.72}$ /carbon cloth was prepared by a facile in situ solvothermal method [43]. Park [44] and co-workers reported supercapacitor electrodes based on urchin-like $W_{18}O_{49}$ nanostructures, and the 3-D hierarchical formation consists of many radial $W_{18}O_{49}$ nanowires, with a highly uniform diameter of ca. 800 nm (Fig. 2). Mesoporous hollow $W_{18}O_{49}$ sphere can be prepared by WCl_6 and acetic acid through solvothermal treatment [45]. The diameter is ca. 600 nm with highly uniform size. It can be seen that these spheres are constructed by bundles composed of rodlike pieces with an average length of ca. 20 nm from the high magnification SEM image. Based on a great deal of experiment data, it can be concluded through lots of experimental results that a low precursor concentration ($C_{WCl_6} \leq 0.5$ g/100 mL) is conducive to the formation of the ultrathin $W_{18}O_{49}$ nanowires, while higher than 1.5 g/100 mL, urchin-like were obtained [46].

Besides the results above mentioned it could be controllable or tunable to the nanocrystal morphology by adding the capping or chelating agent into the solvent. During the reaction, some additives may change the morphology of the tungsten suboxide nanostructures. Liu [47] and co-workers reported synthesis of ordering hydrophilic $W_{18}O_{49}$ nanowires in the presence of PVP coating. Likewise, Jianjian

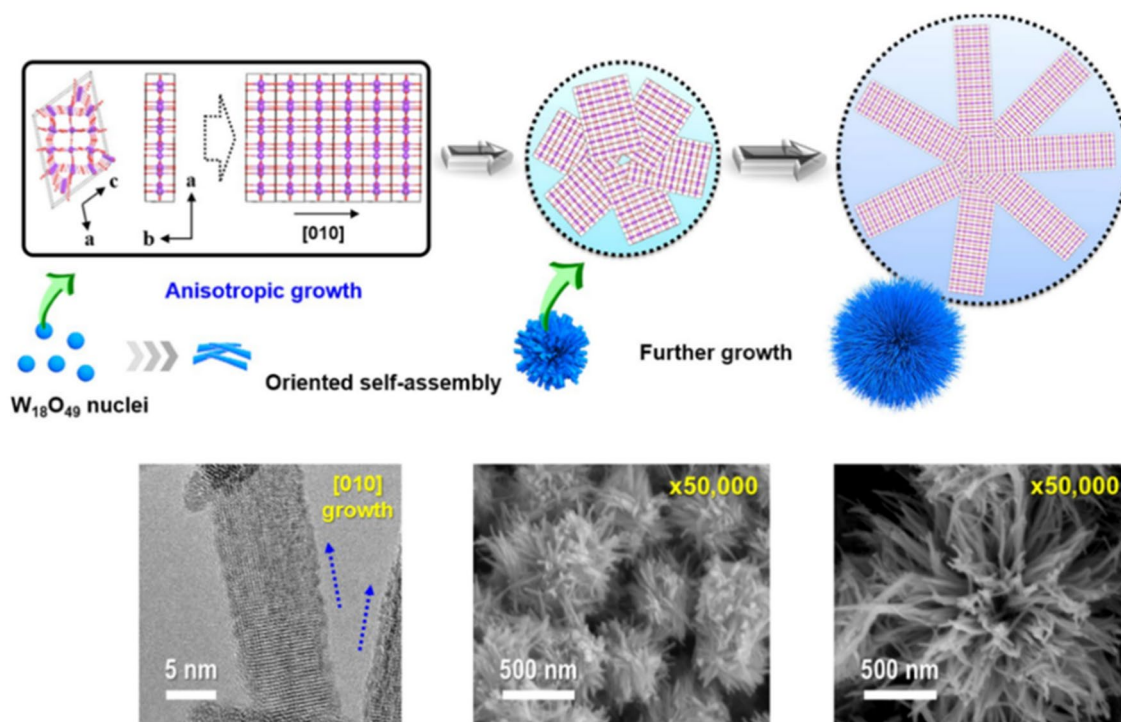


Fig. 2 Schematic diagram of different periods of possible formation of Urchin- $W_{18}O_{49}$. TEM (left) and FE-SEM (center and right) Reproduce with permission [44]. Copyright 2016 Tsinghua University Press and Springer-Verlag Berlin Heidelberg

Qiu [48] synthesized ultrathin PVP-decorated tungsten oxide ($W_{18}O_{49}$) nanowires using a modified solvothermal approach. TEM images showed that $W_{18}O_{49}$ -PVP nanowires possessed excellent microscopic dispersity in aqueous solutions, in contrast to using polyethylene glycol (PEG-400) as ligands. Ultrathin PEGylated $W_{18}O_{49}$ nanowires (average size: length ≈ 50 nm, thickness ≈ 1 nm, width ≈ 4 nm.) have been prepared with ethanol and PEG-400 (volume ratio of 7:3) [49]. When the composition volume ratio of mixture was 4:6, or without PEG-400, the length of nanowires was increased, but the thickness showed no variation [50]. Unique morphology can be fabricated when mixing metal salt into the precursor containing tungsten raw material and solvent. Li [51] have manifested a facile self-assembly route for the $WO_{3-\delta}$ hierarchical made up of monodisperse 3D urchin-like structures indicated by field emission scanning electron microscopy (FE-SEM) examination. The size of $WO_{3-\delta}$ ranges from 700 nm to 1.4 μm . Interestingly, it was found that no self-assembled hierarchical nanostructures are obtained with no Zn^{2+} .

Apart from the absolute ethanol, some other alcohols also can be the solvent in the solvothermal. It already has been released in previous articles that methanol, propanol, cyclohexanol, etc., could be applied to synthesis $W_{18}O_{49}$. Likewise nanowire arrays of $W_{18}O_{49}$ were directly prepared on bare FTO glass [52]. The nonstoichiometric tungsten was

synthesized with WCl_6 as precursor and methanol as solvent. And in a typical procedure of pristine $W_{18}O_{49}$ [53], it can be clearly seen that pure $W_{18}O_{49}$ products consist of compact spindle-shaped particles with size about 150–200 nm when employing n-propanol as solvent. Zhou [54] reported the preparation of hierarchical architecture $WO_{2.72}$ nanorod bundles (NRBs) with propyl alcohol. $WO_{2.72}$ NRBs are highly uniform (ca. 250 nm in length). Similarly, the resulting $W_{18}O_{49}$ nanomaterials possess uniform nanorods bundled and the ends of bundles are packed together loosely [55]. Likewise, the morphology of tungsten suboxide is observed to be homogeneous particles owing thicker central part and thinner ends [56]. Zhen-Feng Huang [57] developed a “one-pot” solvothermal strategy for $W_{18}O_{49}$ NW alignments using NO_3^- -mediated assembly (with the addition of $NaNO_3$). It shows that $W_{18}O_{49}$ is in the form of a spindle-like NW. Tungsten oxide was prepared with WCl_6 and cyclohexanol [58]. The tungsten oxide bundles consist of ultrathin nanowires (diameters: 2–5 nm). The morphology of nanomaterials could be differed via the procedure of synthesis. With little modification, based on the HR-TEM images, it can be confirmed the ultrafine feature of several overlapped individual nanowires [59]. Cheng [60] reported the preparation of ultrathin $W_{18}O_{49}$ nanowires by the Langmuir–Blodgett (LB) technique, the length of which reaches up to several 100 nm, so that the aspect ratio is more than 100.

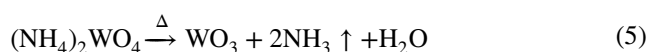
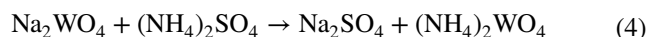
Except for “one-pot” stratagem of WO_x preparation, it has been proposed that two steps involve synthesis of tungsten acid precursor and subsequently the solvothermal reaction is available for the preparation of $\text{W}_{18}\text{O}_{49}$ nanostructures. Lou [61] and co-workers described an inorganic route for synthesis of $\text{W}_{18}\text{O}_{49}$ nanorods. TEM results reveal that the diameters of monodisperse nanorods are 6.9 ± 1.6 nm for 2 h reaction period and 7.3 ± 2.1 nm for 8 h. Hang Chen [62] published a communication about the synthesis of $\text{W}_{18}\text{O}_{49}$ nanowire networks with $\text{Na}_2\text{WO}_4 \cdot 2\text{H}_2\text{O}$ as tungsten resource, and p-amino-benzoic acid (PABA) was used as a reductive and structure-directing agent. The amount of PABA plays an important part during the reaction.

WCl_4 also can be the raw material for the synthesis of nonstoichiometric tungsten oxide [63]. It can be seen that the $\text{W}_{18}\text{O}_{49}$ mainly consists of many homogenous nano-needles, the length of which are about 2 μm , and diameters are in the range of several tens nanometers.

3.2 Vapor-phase synthesis

Vapor-phase synthesis, such as chemical vapor deposition (CVD) and physical vapor deposition (PVD), is scalable and versatile for synthesis oxygen-deficient monoclinic $\text{W}_{18}\text{O}_{49}$ phase as well as WO_3 . Generally, it involves the condensation of targeted material onto a substrate, which is evaporated by many techniques. Thangala [64] and co-workers reported the utilization of CVD to large-scale synthesis of tungsten oxide nanowires and their arrays on a variety of substrates, e.g., amorphous quartz and fluorinated tin oxide. Xianghui Zhang [65] indicated tungsten oxide nanowire arrays by a catalyst-free, high-temperature, thermal evaporation process. By adjusting the parameters, various nanostructures including $\text{W}_{18}\text{O}_{49}$ and WO_3 nanowire networks were obtained. Zhou et al. [66] have successfully synthesized three-dimension tungsten oxide nanowire networks with widths range from 10 to 180 nm, using a thermal evaporation approach, employing W powders in the presence of oxygen. The $\text{WO}_{3-\delta}$ nanowires have a cubic structure, from the TEM investigation and X-ray diffraction (XRD) results. Yubao Li [67] and co-workers have developed a simple route using infrared irradiation to heat W foils under different vacuum condition to prepare two $\text{W}_{18}\text{O}_{49}$ nanostructures, i.e., single-crystalline nanotubes or nanowires. The XRD pattern of nanotube film shows that all peaks can be well indexed to monoclinic $\text{W}_{18}\text{O}_{49}$ phase. And the diameters of nanowires are in the range of 20–100 nm. Zhu [68] succeeded in generating novel treelike microstructures, consisting mainly of WO_x ($x=0-3$) micro- or nano-needles, with ca. 5–50 nm wide and ca. 20–200 nm long. Kunquan Hong [69] reported a thermal evaporation method for fabricating tungsten oxide nanowires at low temperature (~ 400 °C). $\text{W}_{18}\text{O}_{49}$ nanowires with diameters of 10–50 nm, extending to 500–1500 nm

length, are produced with high yield. The cross-sectional shapes are circular or polygonal. Hu [70] described the generation of hollow fibers of crystalline $\text{W}_{18}\text{O}_{49}$, organized in pine-tree-like arrays, by oxidation of WS_2 . An in situ SEM showed that it was arranged by the needles which consist of long straight hollow fibers (3–10 mm outer diameter, 0.1–2 mm in length). And $\text{W}_{18}\text{O}_{49}$ nanorods were obtained by reductive pyrolysis of $(\text{NH}_4)_x\text{WO}_{3+x/2}$ precursors [71]. In a typical procedure, the $\text{W}_{18}\text{O}_{49}$ was prepared by reducing the as-obtained precursors in an H_2 (5 vol%)/ N_2 atmosphere. Likewise, $\text{W}_{18}\text{O}_{49}$ nanorods have been controlledly prepared by pyrolyzing and reducing the $(\text{NH}_4)_2\text{WO}_4$ in 5% H_2 + 95% Ar process [72]. As it is shown in Fig. 3, the reaction involved in the process is shown as follows:



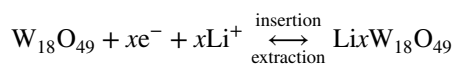
4 Applications

4.1 Electrochemistry

4.1.1 Electrochromic

Electrochromic (EC) is a reversible dynamically color change in a material caused by applied external potential. As one of the paramount branches of chromism, it exhibits many advantages, such as outstanding reversibility, low driving voltages, fast switching speed, and multiple color changes. As a well-known electrochromic material, WO_x ($x \leq 3$) can display colorless and blue by alternately applying positive and negative voltages, which Deb. [73] has propelled to the forefront in investigations of EC. Electrochromic devices are widely studied for use in energy-saving smart windows and antiglare mirrors. The utilization of nanomaterials can supply electrochromic devices with improved coloration efficiency, faster switching time, and potentially reduced costs.

Yu [47] and co-workers prepared an ordered $\text{W}_{18}\text{O}_{49}$ nanowire assembling film by Langmuir–Blodgett technique. The results show coloration/bleaching times of ordered monolayer thickness of film, with good stability (Fig. 4). The mechanism of $\text{W}_{18}\text{O}_{49}$ in 0.1 mol/L LiClO_4 electrolyte can be expressed as



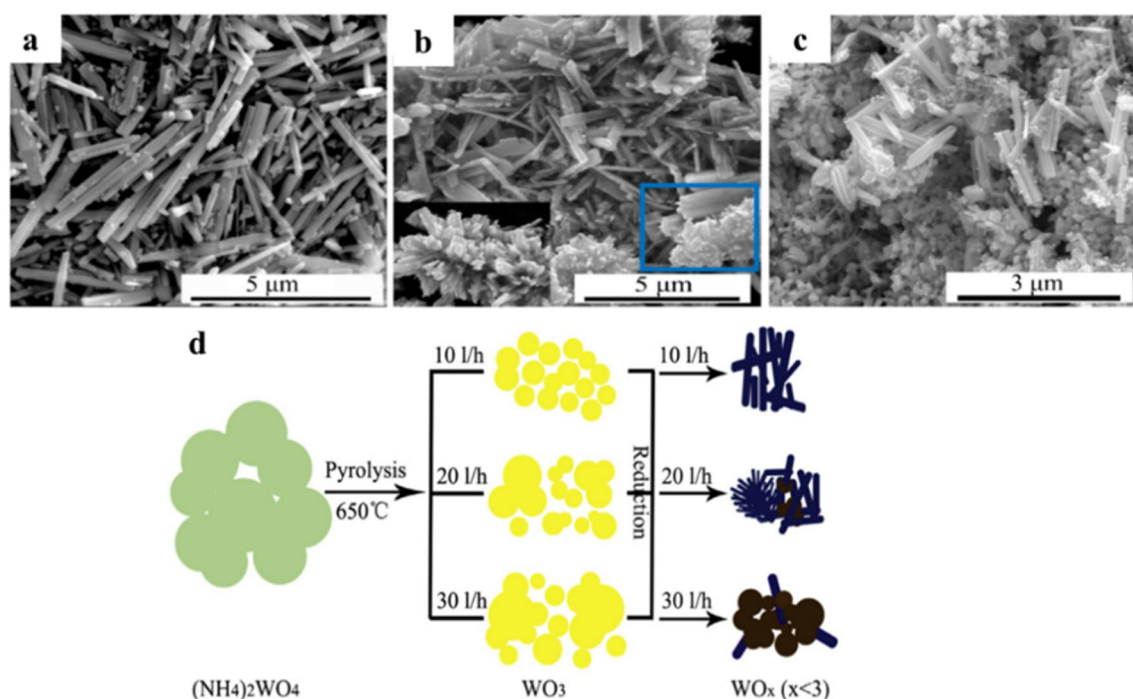


Fig. 3 SEM images **a** 10 l/h, **b** 20 l/h, and **c** 30 l/h. **d** Schematic diagram of rod, flower, and grain formation Reproduce with permission [72]. Copyright 2018 Elsevier B.V. All rights reserved

And $W_{18}O_{49}$ bilayer films were fabricated by using citric acid as a chelating agent [74]. As for EC performance, it exhibited an enhanced property: optical modulation (68.7%) with stability over 400 cycles; fast response times (1.4 s/2.3 s for bleaching/coloring).

Recent discussions regarding electrochromic device have dominated research is that if it is capable of selectively controlling the transmitted sunlight in the VIS and NIR regions. It demonstrated that the WO_x NWs, as a simple single-component material, not only achieve above aims, but also have a high optical modulation of the full solar spectrum toward other nanocomposites. Zhang et al. [29] presented a Al^{3+} intercalation/extraction applied for electrochromic smart windows instead of monovalent ions (e.g., H^+ , Li^+). As a result, the EC smart window delivers impressively well EC performance: high optical modulation (93.2%, 91.7%, 88.5%, at 633, 800, 1200 nm, respectively), fast switching times (8/5 s at 1200 for coloration/bleaching; 16/13 s at 633 nm), and high cyclability (5.5% capacity loss after 2000 cycles).

However, there has been dramatically increasing interest in developing flexible devices for various applications [75]. Heo [21] exemplified a new method to fabricate uniform, mesoporous thin films of $WO_{2.72}$ nanorods. And by introducing guest components (niobium polyoxometalate clusters), a WO_x - NbO_x composite film is produced and that the system exhibits rapid switching response and can be stably cycled

more than 2000 times. Li [76] constructed the mechanically and electrochemically stable bilayer nanowire networks composed of AgNWs and $W_{18}O_{49}$ NWs on an ultrathin polyethylene terephthalate (PET) substrate. These films exhibit enhanced electrochemical and structural stability after 500 bending or 1000 folding cycles.

4.1.2 Supercapacitors

Supercapacitors are deemed to be much promising means of energy storage due to their superiority of high power densities and cycle lives, as well as security and reliable. Yoon et al. [77] synthesized ordered mesoporous WO_x ($x < 3$) which was employed as an electrode for pseudocapacitors exhibits a high-rate capability and excellent volumetric capacitance ($366 \mu F cm^{-2}$, $639 F cm^{-3}$), due to high electrical conductivity ($1.76 S cm^{-1}$) and a large surface area ($54.3 m^2 g^{-1}$). Wang D [78] reported hexagonal mesocrystals formed by the ordered self-assembly $W_{18}O_{49}$ nanowires achieved a specific capacitance of $579 F g^{-1}$ at a scan rate of $20 mV s^{-1}$ in 0.5 M H_2SO_4 . Park [44] reported 3-D urchin-like $W_{18}O_{49}$ nanostructures, providing a large specific surface area as well as mesoporous structure. The electrodes not only show a high specific capacitance of $235 F g^{-1}$ at a current density of $20 A g^{-1}$, but also superior long-life performance. Li [35] prepared a composite electrode consisting of interpenetrating nanonetworks by $W_{18}O_{49}$ nanowires and

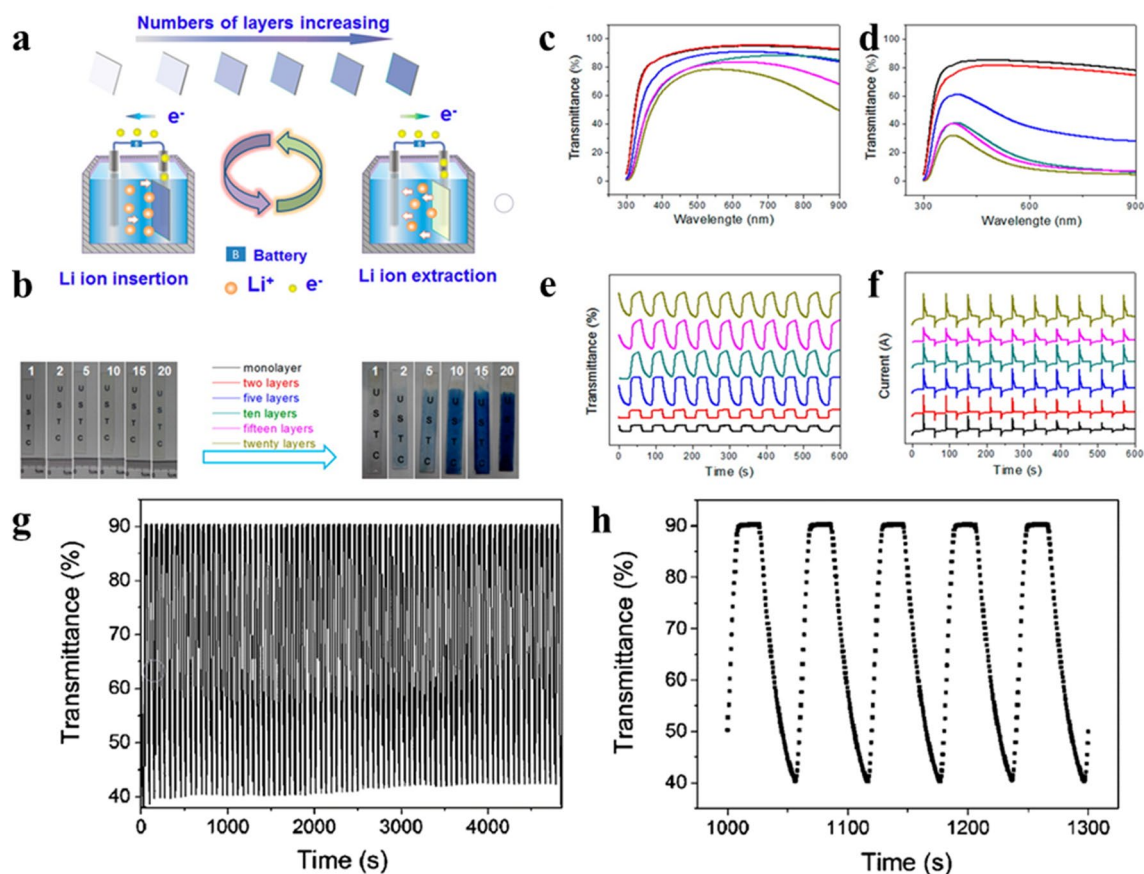


Fig. 4 Electrochromic performance of ordered $W_{18}O_{49}$ nanowire films. **a** Illustration of electrochromic device. **b** Picture of films (left) bleaching and (right) coloration, and **c, d** Optical transmittance of films before (left) and after (right) to application of the external volt-

age. **e, f** Optical transmittance and electrochromic switching of films. **g, h** Stability of electrochromic switching of films. Reproduce with permission [47]. Copyright 2013 American Chemical Society

single-walled carbon nanotubes. These highly connected nanowires and nanotubes possess high conductivity of electrodes (1626 S cm^{-1}) for fast electron transfer. And it exhibits high areal (1.11 F cm^{-2}) and volumetric capacitances (459 F cm^{-3}) at a current density of 2 mA cm^{-2} . Furthermore, Li K [79] recently fabricated a highly flexible composite film, based on Ag nanowires/ $W_{18}O_{49}$ NWs bilayer networks, with electrochromic/actuating (238° bending angle) dual-responsive phenomena. The actuating mechanism for macroscopic deformation is elaborated and further verified as the pseudocapacitive EC-based reversible lattice contraction/recovery of $W_{18}O_{49}$ nanowires during lithium ion intercalation/de-intercalation processes.

Meanwhile, the fabrication of novel supercapacitors devices equipped with new features or smart functionalities is the key to the next generation of technology. Zhao et al. [42] synthesized $W_{18}O_{49}$ and polyaniline for supercapacitor electrodes integrated with the ability of energy display due to synergy effect between each other. Meantime it enables to work in a broadened window ($-0.5 \text{ V} \sim 0 \text{ V}$ for $W_{18}O_{49}$

and $0 \text{ V} \sim 0.8 \text{ V}$ for polyaniline). Similarly, a complementary EC device combining $W_{18}O_{49}$ with Prussian blue not only shows a higher optical contrast (59.05% at 632.8 nm) but also a faster switching response (coloration time: 6.9 s and bleaching time: 1.2 s), superior to the single layer $W_{18}O_{49}$ nanowire device [52].

4.2 Photocatalysis

The photoelectrochemical conversion and storage of solar energy have attracted considerable attention over the last decades [80]. Titanium dioxide (TiO_2) is the most extensively investigated as metal oxide semiconductor and highly efficient photocatalyst up to now [81, 82]. To facilitate further advancement in the field of photocatalysis, it is necessary to develop visible-light-driven photocatalysts [83]. In recent years, tungsten oxide is a wide-bandgap n-type semiconductor, and the advantage of WO_3 is that it can be irradiated by the blue region of the visible light, as its bandgap is in the range of 2.6 to 3.2 eV , depending on the morphology

and stoichiometric properties. Furthermore, it is remarkably stable in acid matrix, making it a promising candidate for treatment of contaminated water [84]. Substantial efforts have been focused on promoting the photocatalytic property of WO_3 , including selective doping, construction of hybrid structures, and synthesis of nanostructured WO_3 [40].

Recently, metallic oxides with oxygen vacancies have gained much importance [85]. Herein, it was reported WO_x nanosheets with abundant vacancies as robust photocatalysts for solar light harvesting. Oxygen vacancies serve as shallow donors in metal oxides such as TiO_2 , WO_3 , and $\alpha\text{-Fe}_2\text{O}_3$, thus facilitating charge transfer between the metal oxide and the electrolyte [84]. It has been reported through an in situ redox reaction between $\text{WO}_{2.72}$ support and metal salt solutions, a series of noble metal/ WO_3 hybrid nanocomposites with various morphologies and sizes were obtained [86].

In recent years, nonstoichiometric $\text{W}_{18}\text{O}_{49}$ has been proposed as a promising photocatalyst owing to its unique defect state and intense near-infrared absorption. The charge transfer mechanism is as below: According to previous reports, the oxygen deficiency shifts the conduction band minimum to a more negative potential than stoichiometric WO_3 . As a result, the conduction band edge of reduced WO_x is negative enough to achieve proton reduction and generate H_2 . Under light irradiation, the charge carriers (i.e., photoexcited electrons and holes) in $\text{W}_{18}\text{O}_{49}$ are reactive toward O_2 and H_2O , respectively, leading to the formation of reactive species such as HO_2 , H_2O_2 , and $\cdot\text{OH}$ through single-electron or multi-electron reduction process. It is generally considered that the conduction band (CB) should be more negative than the potential for the single-electron reduction of oxygen ($\text{O}_2 + \text{e}^- \rightarrow \text{O}_2^- (\text{aq})$, -0.284 V vs NHE; $\text{O}_2 + \text{H}^+ + \text{e}^- \rightarrow \text{HO}_2 (\text{aq})$, -0.046 V vs NHE) to allow efficient consumption of

photoexcited electrons and subsequent oxidative decomposition of organic compounds by holes.

However, the efficiency of photocatalysts is relatively low. Therefore, breaking the balance between light harvesting and charge separation, along with the stability, is crucial for the development of narrow-bandgap photocatalysts.

4.2.1 Reduction of carbon dioxide

Xi [46] investigated the photochemical activity of ultrathin $\text{W}_{18}\text{O}_{49}$ nanowires, which show an superior capability of reduction of carbon dioxide. Generally, the oxygen vacancies would inactivate after photocatalysis reaction. Fortunately, after employing the strong reducing agent of NaBH_4 , it found that the activity could be recovered.

4.2.2 Pollutant degradation

The photocatalytic degradation of toxic pollutants is of great significance and also represents a commonly approach to evaluate the activity of photocatalysts [81]. Incorporation of carbon nanostructures is presented to enhance the photocatalytic activity. The $\text{W}_{18}\text{O}_{49}$ -reduced graphene oxide (RGO) aerogel exhibited good photocatalytic activity and stability over 30 cycles, and six different dyes could be degraded completely within 25 min under visible light irradiation (Fig. 5). The good photocatalytic activity of pure $\text{W}_{18}\text{O}_{49}$ nanowire may arise from their large specific surface, low bandgap of 2.8 eV, and its defect structure. Manipulating and modulating the stoichiometric ratio of semiconductor materials is emerging to extend the range of solar light harvesting, since it can increase the free-charge density and lead to the LSPR that result from the excess free-charges

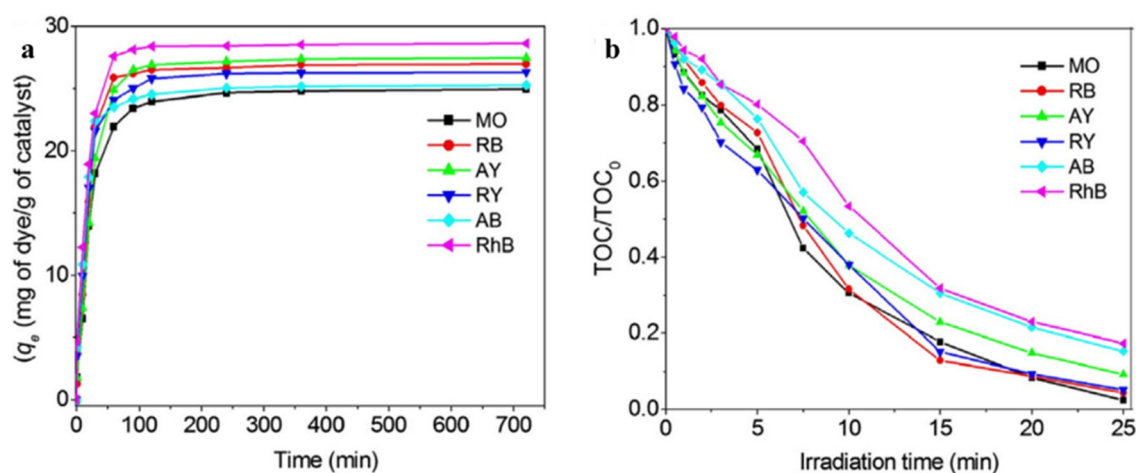


Fig. 5 **a** Adsorption capacity of various dyes with treatment of the $\text{W}_{18}\text{O}_{49}$ -RGO, **b** Total organic carbon (TOC) removal during photocatalytic of various dyes in the presence of heterostructure

$\text{W}_{18}\text{O}_{49}$ -RGO aerogel Reproduce with permission [81]. Copyright 2014 Elsevier Ltd. All rights reserved

on the surface of semiconductor surface. Zhang [87] and co-workers have demonstrated the UV–vis–NIR-driven photocatalytic water reduction prepared with integration of $W_{18}O_{49}$ nanograsses and graphitic carbon nitride ($g-C_3N_4$) nanosheets. The heterostructure showed a 3.2-fold improvement on H_2 evolution as compared to the single $g-C_3N_4$ in spite of the photocatalytically inert $W_{18}O_{49}$. The blue-colored $W_{18}O_{49}$ nanograsses show an intrinsic absorption edge around 410 nm, which is close to the bandgap of stoichiometric tungsten oxide (≈ 3.0 eV), illustrating its semi-conductive property.

4.3 Near-infrared shielding and photothermal therapy

Near-infrared light (NIR) is about spectrum wave band in the range of 780–2500 nm in general. As is well known, it accounts for about 50% of the total solar radiation spectrum, and thus making the maximum utilization of NIR light has been a hot issue with no doubt. The development of an excellent NIR absorbent or controllable unit is an effective way to realize this. Based on the former aspect, various therapeutic strategies, for example, photothermal therapy, photodynamic therapy, are under intense research for cancer so far. As for the latter one, there has been a sharp requirement to shield the near-infrared radiation (or called heat rays) by employing transparent coating on the windows to save the energy and thereby decrease the emission of carbon dioxide.

As revealed by the absorption result, the $W_{18}O_{49}$ nanocrystals show strong and wide absorption in 780–2500 nm [63]. Furthermore, $W_{18}O_{49}$ nanocrystals with controlled dimensions and morphologies (nanorods, nanofibers, nanospheres, nanoparticles, and even nanoassemblies) (Fig. 6) and the optical properties were also investigated systematically [88]. The absorption spectra of the $W_{18}O_{49}$ nanosheets and $Pd/W_{18}O_{49}$ hybrids were compared. Although the noble metals, black compounds, etc., have been reported as NIR shielding candidates, the $Pd/W_{18}O_{49}$ hybrids show low absorption in the visible and NIR region. It can be seen that the $W_{18}O_{49}$ nanosheets also show a broad absorption in the visible and NIR region, and strong absorption in the UV region due to band transition [13].

Phototherapy has remarkable modified selectivity and fewer side effects as compared to traditional radiotherapies and chemotherapies [89]. Photodynamic therapy (PDT) and photothermal therapy (PTT) are two main phototherapeutic approaches, which require absorption of light by a photosensitizer/reagent to generate reactive oxygen species (ROS) and local hyperthermia leading to the death of cancer cells, respectively. An indocyanine green (ICG) attached $W_{18}O_{49}$ nanoplateform has achieved a synergistic effect of two methods on tumor death. In this consideration, $W_{18}O_{49}$ nanostructures work as not only an executed

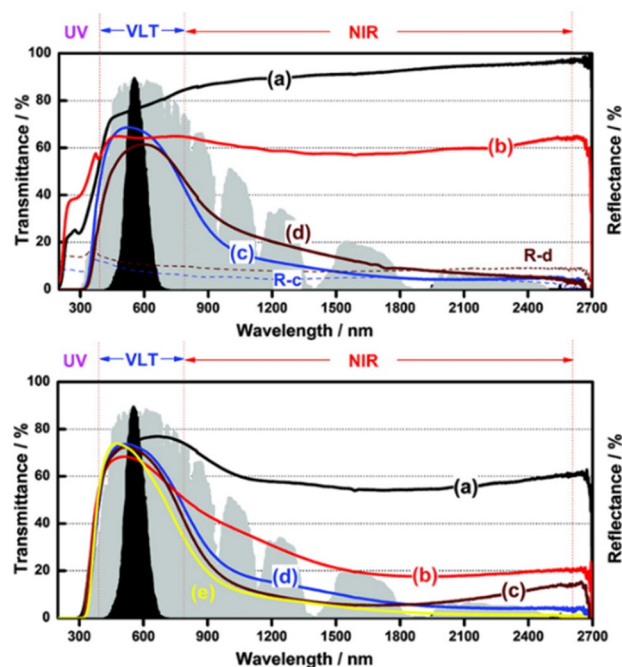


Fig. 6 Above: transmittance spectra of the (a) WO_3 , (b) bulk $W_{18}O_{49}$, and (c) $W_{18}O_{49}$ nanofibers and reflectance spectra, R-c (d) spherical particles and corresponding reflectance spectra R-d. Below: Transmittance spectra of (a) $W_{18}O_{49}$ microspheres: 0.5–2 μm , (b) nanospheres: 200–500 nm and (c) about 80 nm, (d) nanorods: length 40–80 nm and (e) shorter nanorods. Reproduce with permission [88]. Copyright 2012, American Chemical Society

photothermal agent, but also a feasible nanocarrier for photosensitizers due to its hierarchical unique nanostructures. In previous reports [50], under a wavelength of 980 nm laser (0.72 W cm^{-2}), the temperature of the nanowire aqueous dispersion ($0.25\text{--}3.0 \text{ mg mL}^{-1}$) can increase by $12.2\text{--}41.2^\circ\text{C}$ in 5 min, as a result of high photothermal conversion efficiency.

It was reported the ability of PEG- $W_{18}O_{49}$ NWs to sensitize the formation of singlet oxygen upon irradiation with NIR light [49], the existence of the PDT and PTT pathways was proved. The extinction coefficients of PEG- $W_{18}O_{49}$ NWs, which are approximately $10^7 \text{ M}^{-1} \text{ cm}^{-1}$ (e.g., $0.55 \times 10^7 \text{ M}^{-1} \text{ cm}^{-1}$ at 980 nm), are about 2–3 orders of magnitude higher than that of conventional photosensitizers. And the mouse injected with $WO_{3-\delta}$ nanodots showed a temperature increase (within 10 min increment up to 25°C) at the tumor site under irradiation of 808 nm laser [48].

4.4 Sensor

Metal oxides, e.g., SnO_2 , TiO_2 , and WO_3 , have been widely studied for detecting toxic gases. The principle in gas sensing is on the basis of the effect on the resistance when exposing to the gases. Since the reactions are mainly taken place on the surfaces of the materials, to reduce the grain/

particle size or increase the specific surface area are effective approaches to amplify the signals [90]. Another strategy is adding noble metal nanoparticles to the surface, which is known to improve not only sensitivity and stability but also selectivity.

The sensing principle is due to chemical adsorptions and catalytic reactions of gas species on the surface [91]. Thus, the property of metal oxide semiconductor (MOS) gas sensors is dependent on the type of semiconductor and surface morphology. Tungsten oxide (WO_3), an n-type semiconductor, exhibits particularly promising gas-sensing performance toward the detection of NO_2 , NH_3 , O_3 , and H_2 [92]. Besides WO_3 , oxygen-deficient tungsten oxide is also sensitive to gas. Kim et al. reported fabrication of a metal oxide semiconductor gas sensor, as well as its unusual gas-sensing behavior [92]. It shows highly sensitive sensing ability for various reducing and oxidizing analytes even at room temperature. In another article, the nanowires show a quick response (19 s to 5 ppm NO_2) [93]. Compared to WO_3 nanosheets, $\text{W}_{18}\text{O}_{49}$ nanowires showed quicker response–recovery and higher response due to the high specific surface area and the nonstoichiometric crystallology. Cheng [60] and co-workers synthesized single-crystalline ultrathin $\text{W}_{18}\text{O}_{49}$ nanowires with high aspect ratio. The sensor, composed of ten layers aligned nanowires, assembled by Langmuir–Blodgett technique, exhibited excellent room-temperature sensitivity to H_2 in humid air. Zhao [58] and co-workers described the room temperature ammonia sensing properties of ultrathin $\text{W}_{18}\text{O}_{49}$ nanowire bundles. It is found that $\text{W}_{18}\text{O}_{49}$ nanowires transferred from n-type to p-type when the concentration of ammonia was down to sub-ppm level. Recent efforts have been turned toward surface decoration of $\text{W}_{18}\text{O}_{49}$, which significantly improve the photoactivity [94]. Ag/AgCl nanoparticle decorated $\text{W}_{18}\text{O}_{49}$ nanorods were prepared; the gas-sensing properties were apparently enhanced toward several reducing gases, with high sensitivities. And it has been demonstrated self-heated hydrogen gas sensors based on Pt-coated $\text{W}_{18}\text{O}_{49}$ nanowire networks have exhibited attractive performances, e.g., high sensitivity (able to detect down to 50 ppm of H_2), good selectivity (poor response to ethanol, CH_4 , CO, and C_3H_8) [95]. Pt nanoparticles on the surface of the nanowires could further enhance the gas-sensing capability.

5 Conclusion and perspectives

We have presented an overview of the structure of $\text{W}_n\text{O}_{3n-2}$, and $\text{W}_n\text{O}_{3n-1}$, the oxygen-deficient Magnéli phases. Among them only monoclinic $\text{W}_{18}\text{O}_{49}$ was reported to be observed in a distinct form, so we mainly pay attention to the pure $\text{W}_{18}\text{O}_{49}$ phase. Then, a broad highlight of synthesis and the obtained material characteristics have been surveyed and

discussed. Many possible applications like for electrochromic, supercapacitors, sensors, and phototherapy have been developed using this material. Single $\text{W}_{18}\text{O}_{49}$ still encounter some inevitable issues: inclining to oxidize or its unstability. Generally, it combines with noble metals, semiconductors, or carbon material to obtain the hybrid or composite materials, which exhibit superior performance.

Integrating more functionality or novel features into one device is in pursuit of their extensive application areas. Yet it would demand of much sophisticated device architecture that requires multifunctional novel materials, refined processing techniques, and a comprehensive and thorough understanding of the fundamental mechanisms. Its already demonstrated advantages in above applications can endow them with distinctive functions, contributing to spark the innovation of energy-saving technologies in the future. Meanwhile, the cost, long-term stability, and practicability should be considered into the whole procedure.

It is noteworthy that most of these are generally based on rigid substrates such as conducting oxides on glass, e.g., ITO, FTO, and aluminum-doped zinc oxide (AZO) et al. There has been exceeding interest in developing flexible devices, it is still a major challenge to the diverse energy-related applications merged with other smart multifunctionalities, while keeping configurational integrity against mechanical deformation. Besides the above aspects, one of the most promising multifunctional devices are those simultaneously capturing optical modulation performance and energy applications, such as energy storage functions (supercapacitor and battery) with electrochromic into one device.

The application of WO_x may be further extended, from simple laboratory process to device-oriented design and control. Not only this material, $\text{W}_{18}\text{O}_{49}$, but some other new substance may be exploited in multifunctional devices. So it needs to promote the rapid progress of material for the next generation of futuristic technology.

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